

tmux documentation

CONTINUATION.md — vasic-digital tmux

Last updated: 2026-05-21T05:30Z

§0 — How to resume work in any CLI agent

Paste this prompt:

Read CONTINUATION.md at the repo root. Identify the topmost item under §3 Active work with status IN PROGRESS or BLOCKED. Re-read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md for mandates and current backlog. Resume from current state. Update this document as you work.

§1 — Snapshot

Field	Value
Repo	vasic-digital/tmux on GitHub + GitLab
Origin	Migrated from ATMOSphere project (scripts/tmux/, docker/ Dockerfile.tmux-build, docs/guides/ TMUX_OPTIMIZED_BUILD.md) on 2026-05-07
Pinned tmux	upstream tag 3.6a
Version	1.0.4 (versionCode 5) — CodeGraph (§11.4.78) + verbatim covenant propagation + AUDIT-1/2 fixes + docs reorganisation, 2026-05-21 bash scripts/setup.sh --verify-only → GREEN; suite PASS=18 FAIL=0 SKIP=5 (Linux-only/destructive SKIPS — same profile as prior releases). NEW tests: 19 covenant propagation (PASS=7/0/0), 20 codegraph installed Verification (this cycle) (PASS=5/0/0), 21 codegraph index present (PASS=4/0/0), 22 codegraph MCP wired (PASS=7/0/0), 23 tmx kill shorthand (PASS=5/0/0). Meta-test 26 caught / 0 escaped / 6 skipped (6 SKIPS §11.4.3 topology-correct — Linux-only mutations). e2e PASS=9/0/0. CodeGraph index 6 nodes (small per honest §11.4.6 gap — shell parser not in CodeGraph 0.6.8). All captured 2026-05-21 on Darwin arm64. constitution/ submodule (HelixConstitution, pinned 7f738df) — universal Governance base; project Constitution.md (Project Articles §101–§109, extends the docs submodule), CLAUDE.md, AGENTS.md, QWEN.md, Issues.md, Fixed.md, this document

§2 — Mandates (canonical authority)

- Constitution.md §1 anti-bluff covenant (verbatim user-mandate quote propagated)
- Constitution.md §11.4.1 FAIL-bluffs equally forbidden
- Constitution.md §11.4.2 recorded-evidence requirement
- Constitution.md §11.4.3 per-host-topology test dispatch
- Constitution.md §11.4.4 test-interrupt-on-discovery + 4-layer test coverage

- `Constitution.md` §11.4.5 audio + video quality analysis (N/A for `tmux` scope; principle generalized)
- `Constitution.md` §11.4.6 — No-guessing mandate (verbatim user-mandate quote)
- `Constitution.md` §9 absolute data safety
- `Constitution.md` §12.6 60% host memory budget (per-session `TMX_MEM` is the enforcement)
- `Constitution.md` §5 / §12.10 continuation-document sacred invariant

§3 — Active work

§3.1 Governance bring-up — `Issues.md` / `Fixed.md` / explicit §11.4.x anchors landed

Status: COMPLETE (2026-05-08T17:06Z).

- Created `Issues.md` (open / in-flight tracker) with:
 - Categories A-E, status conventions OPEN / PARTIAL / BLOCKED / RUNNING / INVESTIGATED
 - Status reclassification rules referencing §11.4.6
 - Seeded items: META-MUT-001, CHAL-COVER-001, TEST-AUDIT-001, TMX-T5, TMX-T7, TMX-T8, TOPO-DISPATCH-001
- Created `Fixed.md` (closed archive) with:
 - A0 initial migration (commit `08d4ba5`, 2026-05-07)
 - B0 §1 covenant propagation (commit `b92bf7f`, 2026-05-08)
 - C0 test 09 crash-isolation-scope landed and green (commit `b92bf7f`, 2026-05-08)
- Updated `Constitution.md`:
 - Explicit §11.4.1, §11.4.2, §11.4.3, §11.4.4, §11.4.5 anchors added (preserving §1 prose verbatim)
 - §11.4.6 no-guessing mandate added with verbatim user-mandate quote
 - §9 absolute data safety added
 - §12.6 60% host memory budget added (`TMX_MEM` enforcement seam)
 - §5 expanded into full §12.10 continuation-document mandate
- Updated `CLAUDE.md` and `AGENTS.md` to carry every numbered anchor + canonical-authority cross-references

§3.2 Per-session containerization (Phase B — multi-session work)

Status: RESEARCH COMPLETE + WRAPPER LANDED + ISOLATION VERIFIED (2026-05-08).

Web research output: `docs/plans/containerization.md` recommends `systemd-run --user --scope (cgroup-v2 transient scope) over podman-per-session`. Wrapper at `scripts/tmx` implements `tmx {new|attach|ls|kill}` with `MemoryMax=$TMX_MEM` (default 8G), `CPUQuota=$TMX_CPU` (default 200%), `TasksMax=4096`, `Delegate=yes`.

Test 09 (`scripts/tests/09_crash_isolation_scope.sh`) — 14 PASS / 0 FAIL / 0 SKIP — verifies T1 (host capability), T2 (wrapper invariants), T3 (cgroup interface evidence), T4 (SIGKILL containment + `user.slice` survival), T6 (concurrent registration independence). Source-line breakdown:

- **T1:** `systemd` 258 + `cgroup v2` mounted ✓

- **T2:** tmx wrapper invokes `systemd-run --user --scope` + sets `MemoryMax/CPUQuota/TasksMax/Delegate=yes` ✓
- **T3:** transient scope created; `/sys/fs/cgroup/.../memory.max` reads 268435456 bytes (matches set 256M) + `/sys/fs/cgroup/.../cpu.max` reads 50000100000 (50% quota over 100ms period) — both POSITIVE EVIDENCE per §1 ✓
- **T4:** spawn scope, read `MainPID` from `cgroup.procs`, SIGKILL it, verify scope inactive AFTER kill, verify `default.target=active` THROUGHOUT (user.slice survives — Constitution §1 invariant) ✓
- **T6:** 3 concurrent scopes, all registered + active simultaneously ✓

§3.3 Pending tests — moved to Issues.md

The previously-listed deferred tests (T5 memory pressure under cap, TasksMax stress, Concurrent OOM independence) have migrated to `Issues.md`:

- **TMX-T5** — `Issues.md` C1 (PARTIAL; operator-unblock runbook for dedicated test host)
- **TMX-T7** — `Issues.md` C2 (OPEN; new test 11 planned)
- **TMX-T8** — `Issues.md` C3 (OPEN; new test 12 planned)

Cross-cutting blockers also tracked in `Issues.md`:

- **META-MUT-001** — `Issues.md` A1 (paired-mutation harness needed before §11.4.4 layer-4 coverage is real)
- **CHAL-COVER-001** — `Issues.md` B1 (HelixQA Challenge entries pending)
- **TOPO-DISPATCH-001** — `Issues.md` D1 (formal topology dispatch matrix)

§3.4 Anti-bluff enforcement — test audit + challenges fix + covenant propagation verification

Status: COMPLETE (2026-05-08T18:00Z).

- Verified anti-bluff covenant propagation across ALL governance files:
 - `Root Constitution.md`: §1 + §11.4.1-§11.4.6 present ✓
 - `Root CLAUDE.md`: full covenant present ✓
 - `Root AGENTS.md`: compact covenant present ✓
 - `Containers/CLAUDE.md`: full covenant present ✓
 - `Containers/AGENTS.md`: full covenant present ✓
 - `tmux/` (upstream submodule pinned to 3.6a): N/A — no governance files, not modified per Constitution
- Performed line-by-line §11.4.2 anti-bluff audit of tests 01-09:
 - All 9 tests confirmed to carry positive runtime evidence (/proc files, cgroup interfaces, tmux command output, systemctl state).
 - No FAIL-bluff vectors found (all `set -uo pipefail`, guarded variables).
 - Full audit table moved to `Fixed.md` B2.
- Fixed broken paths in `scripts/challenges/tmux.yaml` (`scripts/tmux/tests/` → `scripts/tests/`).
- Migrated TEST-AUDIT-001 from `Issues.md` B2 → `Fixed.md` B2.

§3.5 Hostname-derived status-bar colour — algorithm + wrapper + 2 tests + challenges

Status: COMPLETE (2026-05-08T19:00Z).

- Created `scripts/hostname_color.sh` — DJB2 hash over hostname → curated 27-colour palette. Deterministic, testable standalone.
- Updated `scripts/tmux.template` — added `_apply_host_color()` function that invokes `hostname_color.sh` and applies `set -g status-style bg=<colour>` on the running tmux server. Applied on both new-session and attach paths.
- Created `scripts/tests/10_hostname_color_algorithm.sh` — 5 invariants (deterministic, valid colourNNN, palette member, spread $\geq 12/16$, empty-fallback to system hostname). All 5 PASS on this host.
- Created `scripts/tests/11_hostname_color_integration.sh` — verifies wrapper applies correct colour to tmux server. SKIPS if binary/wrapper not yet built.
- Updated `scripts/tests/run_all.sh` — glob changed from `0[1-9]*.sh` to `[0-9][0-9]*.sh` to support tests 10+.
- Updated `scripts/challenges/tmux.yaml` — added TMUX-CH-10 (algorithm) and TMUX-CH-11 (integration).
- Updated `docs/research/customization/colors.md` — added implementation documentation covering algorithm, integration, verification, usage, and anti-bluff covenant.
- Updated `AGENTS.md` — bump test count from 9 to 11.

§3.6 Full coverage — missing challenge entries, destructive tests, topology dispatch

Status: COMPLETE (2026-05-08T20:00Z).

- **CH-09:** Added missing challenge entry for crash isolation scope (test 09) — was the only test without a challenge.
- **Test 12** (`12_memory_pressure_under_cap.sh`): TMX-T5 — allocates up to `MemoryMax+10%` inside a transient scope, captures `dmesg` OOM-kill evidence, validates `user.slice` survival. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 13** (`13_tasksmax_stress.sh`): TMX-T7 — fork-bomb resistance. Spawns processes up to `TasksMax=4096`, reads `pids.current/pids.max` from `cgroup` interface. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 14** (`14_concurrent_oom_independence.sh`): TMX-T8 — three scopes A/B/C, OOM-kills A, verifies B and C survive with original `MainPIDs`. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **CH-12/13/14:** Challenge entries for all three destructive tests.
- **Topology dispatch:** added `_probe_topology()` to `scripts/tmux.template` — detects `systemd` version + `cgroup v2`, classifies host as `tmx-supported/tmx-degraded/tmx-unsupported` per §11.4.3.
- **Fixed.md B2:** closure SHA updated to `68a65b0`.
- **Issues.md** → **Fixed.md migration:** B1 (CHAL-COVER-001), C1 (TMX-T5), C2 (TMX-T7), C3 (TMX-T8), D1 (TOPO-DISPATCH-001) moved to `Fixed.md`.

§3.8 Audit cycle 2026-05-13 — submodule pin-drift caught + governance staleness fixed

Status: COMPLETE (2026-05-13T00:00Z).

Triggering event: user invoked `/init` followed by "what is left unfinished, no-bluff policy heavily enforced everywhere". Audit found and remediated:

- **CRITICAL §1 / §11.4.6 bluff:** `f4132aa` `Auto-commit` (2026-05-13) silently bumped `tmux` submodule from `cc117b5` (tag `3.6a`) to `3f651d9f` (`3.6a-329-g3f651d9f` — 329 commits past the only stable tag). Governance docs across `Constitution / CLAUDE / AGENTS / CONTINUATION / README` still claimed "pinned to tag 3.6a". User decision: revert to tag (newer stable tag does not exist; 329 commits are unreleased upstream master). Submodule pointer reverted to `cc117b5`.
- **README.md staleness:** line 5 ("eight verification tests"), line 33 (`PASS=6 FAIL=0 SKIP=2`), line 37 ("8 tests cover"), line 39 (2-SKIP list) — all rewritten to current 14-test / `PASS=10 SKIP=4 / 4-SKIP-list` state with explicit `TMX_TEST_DESTRUCTIVE=1` callout and §11.4.4 layer-4 harness reference.
- **CLAUDE.md drift:** added top-of-file project summary + canonical-doc cross-links + fresh-conversation workflow; fixed `0N_*.sh` glob → `NN_*.sh`; named the four layers inline in the Test-interrupt rule; added "Files to never edit directly" section (parity with `AGENTS.md`).
- **AGENTS.md drift:** mirrored `CLAUDE.md` project summary + cross-links; fixed same `0N_*.sh` glob bug; added meta-test row to commands table.
- **Issues.md format:** restored `A / B / C / D / E` section headers as empty sections (was: `A / B / E` only, with `C` and `D` missing despite conventions list referencing them). All bodies note "landed in `Fixed.md`" with item IDs for traceability.
- **CONTINUATION.md timestamp:** `2026-05-08T22:00Z` → `2026-05-13T00:00Z`; this entry added under §3.8 per §12.10 invariant.
- **M6 mutation added** (`scripts/tests/meta_test_false_positive_proof.sh`): injects `$$` into the `hostname_color` hash, making the algorithm non-deterministic per-invocation. Test 10 T1 must FAIL under this mutation. Closes the audit-flagged gap: until M6, the "same-host = same-color" user invariant was protected by side-effect only (no dedicated anti-randomness mutation).
- `docs/guide/README.md` **phantom-script bluff fixed:** 3 occurrences each of `verify_tmux.sh` → `verify.sh`, `setup_tmux.sh` → `setup.sh`, `install_tmux_deps.sh` → `install_deps.sh`; "8 tests" → "14 tests" (2×).
- **GUIDE.md "severity hierarchy" pre-existing bluff caught** (`Fixed.md` A3): documented "blockers / critical / advisory" classification does not exist in `run_all.sh` — every test is treated equally (any FAIL = RED). Rewritten to describe the actual gate logic. Caught while extending the table for tests 09-14; would have been silently doubled otherwise.
- **Verification + validation cycle 2026-05-13** (`Fixed.md` A14): operator invoked `superpowers:verification-before-completion` + asked for full anti-bluff verification + governance propagation. Every verification command run FRESH this session per Iron Law. Captured evidence: `setup.sh --verify-only SUMMARY=PASS=10 FAIL=0 SKIP=5 GREEN`; `test_e2e.sh PASS=9/0/0 GREEN`; `test 15 PASS=6/0/0`; `live tmx new -s VerifyDemo` shows operator's host shell (`milosvasic@Mistborn`, which `brew=/opt/homebrew/bin/brew`, `ulimit -t=86400`, `ulimit -u=2666`). Two defects caught and fixed: (D1) wrapper `hostname` resolution drifted to FQDN after bridge removal — restored `scutil-LocalHostName` path so `colour` stays `colour202` for `Mistborn`; (D2) `install_deps.sh` + `build_oom_set.sh` weren't OS-aware out of the box — now `install_deps.sh` detects `Darwin` → `brew install` (no `sudo`), `build_oom_set.sh` SKIPS on non-Linux. Governance propagation completed: verbatim user-mandate quote now in root `CLAUDE.md` + `AGENTS.md` (previously only in `Constitution` + `Containers`); §11.4.7 reference added to `Containers/CLAUDE.md` + `Containers/AGENTS.md` (previously only `Containers/CONSTITUTION.md`). Install state: shell snippets in both `~/.bashrc` + `~/.zshrc`, `zsh -c 'which tmx; tmx -V'` returns native wrapper + `tmux 3.6a`. macOS

RLIMIT_AS gap documented honestly per §1 (XNU EINVAL for unprivileged setrlimit on memory).

- **Per-session isolation + Constitution §11.4.7** (Fixed.md A13): user reported core@localhost + lack of OOM containment. Forensic: `tmx new -s isol1 -d + tmx new -s isol2 -d` previously shared ONE cgroup (`run-p504653-i504654.scope`) — README's "if one session OOMs, others survive" was a §1 bluff. Architectural rewrite: each `tmx new -s NAME` now spawns its OWN tmux server on socket `tmx-NAME` inside its OWN scope `tmx-NAME.scope` with host-adaptive `MemoryMax` ($\max(\text{MemTotal} \times 60\% / 4, 2\text{GB})$), `CPUQuota=200%`, `TasksMax=4096`, `Delegate=yes`. OOM in any session = contained to that scope; B+C survive with original MainPIDs (verified test 14 PASS=8/0/0 with `stress-ng tmx send-keys`). Test 15 NEW: per-session cgroup distinctness (6 assertions, positive readback from `/sys/fs/cgroup`). Test 11 + test 14 + test 08 rewritten to OPERATOR PATH (`tmx new -s NAME` instead of hand-crafted `systemd-run --user --scope`). e2e T7 NEW: bridge confirms two distinct scopes via VM `systemctl`. Meta-test M5 retargeted, M7 retargeted, M9+M10 NEW (10/10 mutations caught). Plan + operator decisions captured in [docs/plans/per-session-isolation.md](#). **Constitution §11.4.7 NEW** (operator-path test coverage rule): every gate test MUST exercise the same entry point an end-user invokes; tests that hand-craft equivalents are supplementary. Propagated to `Containers/CONSTITUTION.md` at same depth. Forensic anchor: Fixed.md A12 (status-bar green default) + A13 (sessions sharing one cgroup) — both shipped while non-operator-path tests reported GREEN.
- **Regression protection for A10 — gate-coverage gap closed** (Fixed.md A11): user demanded "make sure nothing passes again! zero-bluff policy MUST BE followed blindly!" — honest accounting of why the existing gates missed the colour bug:
 - test 11 always passed -S "\$SOCKET" → tested explicit-socket path only; operator's default-socket use case uncovered.
 - meta-test M4/M5 targeted `scripts/tmx` (the SSH bridge on Darwin) instead of `scripts/tmx-vm` (the actual VM wrapper) — paired-mutation harness was mutating the wrong file.
 - test_e2e.sh pre-fix didn't read `status-style` at all.
 - Triple-layer regression protection added: (1) test 11 T6 — new test exercising default-socket path with positive evidence (`tmux show -g status-style` with NO -S, FAILs explicitly on `bg=green` default-applied bluff); (2) meta-test M4/M5 retargeted to `scripts/tmx-vm`; (3) M7 (re-introduce SOCK-empty early return) + M8 (hardcode `bg=green`) both caught by test 11 T4.1/T6. Result: all 8 mutations caught + reverted (16 PASS), full destructive suite PASS=14/0/0, e2e PASS=8/0/0.
- **Status-bar colour silently green (default) + bridge ignored macOS hostname** (Fixed.md A10): user said "Bottom was green for current host. Is that expected for mistborn?" — two stacked bluffs caught:
 - `_apply_host_color/_apply_oom_score` had `[ -n "$sock" ] || return 0` — silently bailed when `tmx new -s` Test was invoked WITHOUT -S (which is the primary operator use case; tmux uses its default socket then). So `set -g status-style` was never fired → tmux default `bg=green` shown. Test 11 always passes -S "\$SOCKET" so the bug was uncovered. Fix: helpers now build `target=(-S "$sock")` only when sock is non-empty; otherwise omit -S and let tmux use its default socket.
 - Bridge `scripts/tmx-mac.template` didn't forward macOS hostname → `hostname_color.sh` inside VM resolved against `localhost.localdomain` regardless of macOS host. Every macOS user got the same VM-derived colour, violating host-distinguishability. Fix: bridge captures `scutil --get LocalHostName` (e.g., "Mistborn"), forwards as `TMX_HOSTNAME=<host>` in the remote SSH command. Wrapper's

- `_apply_host_color` passes `${TMX_HOSTNAME:+"$TMX_HOSTNAME"}` to `hostname_color.sh` — env var is source of truth when set, `$(hostname)` fallback otherwise.
- New `test_e2e.sh` T4.5 explicitly reads `tmx show -g status-style`, computes expected colour from `scutil --get LocalHostName` (or `hostname` on Linux), FAILs on `bg=green` (default-applied bluff) or non-match. Captured runtime evidence on Mistborn: `status-bg 'colour202'` matches `hostname-derived 'colour202'` for 'Mistborn'.
- **Interactive tmx new fixed + e2e automation harness** (Fixed.md A9):
 - User ran `tmx new -s Test` interactively after install and hit `open terminal` failed: `not a terminal`. The bridge + `ssh -t` allocated TTY correctly; the WRAPPER's `cmd & wait` pattern was disconnecting tmux from the foreground process group → no TTY access. Detached-mode tests (`-d`) all passed because tmux daemonizes; interactive (no `-d`) hit the bluff. §11.4.1 FAIL-bluff in the wrapper itself.
 - Fix: wrapper now detects `-d/-D` in args (INTERACTIVE flag), and for interactive: runs `systemd-run` in foreground inheriting TTY, injects `-d` after the new-session keyword (so tmux server detaches), then `exec attach` to take over TTY interactively. For detached: keeps existing path.
 - New: `scripts/test_e2e.sh` — 7-test operator-facing automation: prerequisites + `tmx -V` (via bridge) + new-session + send-keys + capture-pane + survive-without-client + kill-session. All with positive runtime evidence; trap-on-EXIT cleanup. Captured the literal marker string echoed inside the session pane through the bridge — proves the full operator stack carries intent end-to-end.
 - Verified: `bash scripts/test_e2e.sh GREEN 7/0/0`; `TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` still GREEN 14/0/0 (wrapper change didn't regress existing tests).
- **macOS tmx fully operational + side-by-side end-to-end** (Fixed.md A8):
 - `bash scripts/setup.sh` on Darwin now diverges intelligently: builds (containerized), generates BOTH `scripts/tmx-vm` (Linux wrapper for VM paths) AND `scripts/tmx` (Darwin SSH bridge from new `scripts/tmx-mac.template`); runs `scripts/test_vm.sh` for §11.4 target-env verification (since binary cannot `exec` on Darwin); on GREEN, appends `bashrc` snippet to BOTH `~/.bashrc` AND `~/.zshrc` (macOS defaults to `zsh`).
 - **Bridge mechanics:** `scripts/tmx-mac.template` discovers `podman machine`'s SSH endpoint via `podman machine inspect` at every call (port re-discovery — survives `podman machine stop && start`), verifies `tmx-vm` is executable in the VM, then `ssh -t -i <identity> -p <port> core@127.0.0.1 "<vm-repo>/scripts/tmx-vm <quoted-args>"`.
 - **Verified end-to-end on Darwin 24.5.0 / Apple Silicon:** `zsh -c 'source ~/.zshrc; which tmx; tmx -V' → /Users/.../scripts/tmx + tmux 3.6a`; `system /opt/homebrew/bin/tmux -V → tmux 3.6a` — side-by-side coexistence proven; VM-side verify GREEN PASS=11 FAIL=0 SKIP=3.
 - `scripts/verify.sh` **now respects \$WRAPPER and \$TMUX_BIN env vars** so `test_vm.sh` / bridge tooling can redirect to the right wrapper without source modifications.
 - **README.md** gained a new "Architecture" section with ASCII diagram showing Linux-native vs Darwin-bridge dispatch; `docs/guide/README.md §5.5` documents the bridge mechanics + Mermaid flowchart; CLAUDE/AGENTS one-liners updated.
- **Final sweep — env-specific wrapper + §255 violations + sed portability** (Fixed.md A7):
 - **Environment-bound wrapper:** `scripts/tmx` was generated for one env (container / repo paths OR VM / Users / ... / paths) and silently failed in the

- other. New `scripts/test_vm.sh` orchestrator regenerates the wrapper with VM-native paths before each VM-side run. Both `test_containerized.sh` and `test_vm.sh` now own the wrapper-regeneration step for their respective target.
- **Constitution §255 violations:** "ATMOSphere" in `GUIDE.md` title + body, challenge yaml description, `oom_set.c` license comment, `tmux.conf.template` attribution; `atm_tmux_test_*` / `atm_test_*` / `atm_tmux_test_color_*` socket/session names across 8 test files; `~/ .tmux.conf.pre-atmosphere` backup naming. All renamed: code uses `tmx_*` / `pre-vasic-digital`; docs branded "vasic-digital".
 - **sed -i portability:** `setup.sh` used GNU-only `sed -i` for `bashrc` markers — would fail on macOS BSD `sed`. Centralized in `_strip_bashrc_snippet()` using `perl -i -ne ... .. / ... flip-flop range delete` (portable GNU+BSD).
 - **Verified post-fix in VM:** `TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` → `PASS=14 FAIL=0 SKIP=0; META=1 bash scripts/test_vm.sh` → 12/12 mutations caught.
 - **Install-mechanism side-by-side bluffs** (`Fixed.md A6`): user asked about the alias and side-by-side coexistence requirement. Audit found three real defects:
 - `bashrc` snippet pointed `PATH` at `scripts/tmux/` (phantom subdir; wrapper is at `scripts/tmx`).
 - `alias tmux='tmx'` line shadowed the system `tmux` command → broke side-by-side.
 - `setup.sh` closing message said "Then 'tmux' will use the ATMOSphere build" (wrong command name + stale branding).
 - **Canonical answer to user's question:** the alias is `tmx` (not `tmux`). Wrapper at `scripts/tmx` generated from `tmx.template`. `tmx new|attach|ls|kill` invokes the verified `vasic-digital` `tmux`; `tmux` invokes whatever was on the operator's `PATH` before (system / Homebrew / `apt` / `dnf`). Both coexist.
 - **Full destructive + meta-test cycle caught 6 more FAIL-bluffs** (`Fixed.md A5`): pushing past `PASS=10 SKIP=4` into actually running `TMX_TEST_DESTRUCTIVE=1` + meta-test in the podman machine VM (Fedora CoreOS 42 + `systemd 257` + `stress-ng` + `tmx-oom-set` `setcap`) surfaced multiple §11.4.1 `FAIL-bluffs`. Final: **PASS=14 FAIL=0 SKIP=0** (full verify) + **12/12 mutations caught** (meta-test). Real defects fixed:
 - Tests 12/14 `_skip;<continue>` bug — printed `SKIP` but didn't exit; continued running and `FAILED` on missing `stress-ng` → false §11.4.1 `FAIL-bluff`. Fix: `exit 0`.
 - Tests 12/14 unprivileged `dmesg` → fallback to `journalctl -k` via `_kring_count / _kring_tail` helpers; broader OOM regex.
 - Test 13 fork-storm OOM-killed its own scope (`4096 sleeps × 700KB > 256M MemoryMax`); two-phase scope, polling registration, test `TASKS_MAX 4096→256` with `MemoryMax=512M` (production wrapper still `TasksMax=4096` — separately `grep-verified` by test 09 T2.2).
 - Meta-test `sed -i strip-exec-bit` → `orig_mode` capture/restore via `stat + chmod`.
 - Meta-test M5 `sed` pattern was eval-expanded (`${TMX_MEM:-8G}` → `8G`) so it never matched → silent no-op → mutation escaped (a **bluff in the gate itself**). Fix: simplified to literal `MemoryMax=|MemMax=` pattern.
 - Meta-test had no debug visibility → added opt-in `TMX_META_DEBUG=1`.
 - **The user's "same-host = same-color" invariant is now PROVEN end-to-end** with `tmux show -g status-style colour166` on first session AND on second session on same host (test 11 T4.1 + T5).

- **Build pipeline bluffs caught while reproducing on macOS** (Fixed.md A4): three real defects surfaced when user asked why we couldn't build on macOS:
 - `docker/Dockerfile: groupadd -g 20` collided with Ubuntu's dialout group → build aborted at step 7/10. Fix: `-o` (non-unique) on `groupadd + useradd`. Build now reproducible on Darwin hosts.
 - `docker/build_inside_container.sh: LDFLAGS -ljemalloc` was being dropped by linker's default `--as-needed` because `tmux` doesn't reference `jemalloc` symbols. **README's "Build-time -ljemalloc" claim was a §1 bluff until this commit.** Fix: `-Wl,--no-as-needed -ljemalloc -Wl,--as-needed`. Verified `ldd` now shows `libjemalloc.so.2` in `DT_NEEDED`.
 - `docker/build_inside_container.sh: make -j2` silently no-op'd after `LDFLAGS` change (existing binary defeated `mtime` check). Fix: prepend `make clean` so `LDFLAGS` changes always take effect.

The `f4132aa` Auto-commit itself is a §12.10 / §11.4.6 violation — opaque commit message, silent submodule mutation, no CONTINUATION update. Cannot rewrite history per §9 data safety; documenting here is the audit trail.

§3.7 §11.4.4 layer-4 paired-mutation harness landed (A1 META-MUT-001)

Status: COMPLETE (2026-05-08T21:00Z).

- Created `scripts/tests/meta_test_false_positive_proof.sh` — §11.4.4 layer-4 harness with 5 registered mutations:
 - **M1:** break `hostname_color` output format → test 10 T2 FAILs (invalid format)
 - **M2:** force hash to zero → test 10 T3 FAILs (zero spread)
 - **M3:** single-entry palette → test 10 T3 FAILs (palette mismatch)
 - **M4:** remove `systemd-run` flag from `tmx.template` → test 09 T2 FAILs
 - **M5:** remove `Delegate=yes` from `tmx.template` → test 09 T2 FAILs
- Results on this host: **10 PASS / 0 FAIL / 0 SKIP** — all mutations caught.
- Updated `CLAUDE.md`: removed "PENDING" references, wired mandatory operations to the actual file.
- Updated `AGENTS.md`: layer 4 now marked as landed, not PENDING.
- Updated `Fixed.md`: A1 entry with closure details; all "PENDING META-MUT-001" references replaced with actual coverage references.
- Updated `Issues.md`: A1 removed (→ `Fixed.md` A1). `Issues.md` now has **zero open items** — all originally seeded items are closed.
- Cleaned up `CONTINUATION.md` §3.6: removed stale "remaining open" note.

§3.10 — CodeGraph (§11.4.78) + verbatim covenant + AUDIT fixes + docs reorg (2026-05-21)

Status: COMPLETE (2026-05-21T05:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): create full working plan for tackling all these points completely; all existing tests + Challenges MUST work anti-bluff; covenant MUST be part of project + submodule Constitution / CLAUDE / AGENTS; `HelixConstitution` submodule is the root.
- Operator (2026-05-21): incorporate / install `CodeGraph` for `Claude Code`, `OpenCode`, `Kimi CLI`, `Crush`, `Qwen Code`; comprehensive anti-bluff tests.
- Operator (2026-05-21): full documentation + user guides + HTML + PDF exports.
- Operator (2026-05-21): do NOT modify constitution submodule, only keep it regularly updated.

- Operator (2026-05-21): documentation under context-named subdirectories per the constitution rule.

Plan + execution captured at docs/plans/v1.0.4.md. Eight phases all landed in one v1.0.4 commit per §11.4.42 iteration discipline.

- **A18 CodeGraph integration (§11.4.78).** CLI v0.6.8 installed; .codegraph/config.json tracked with §11.4.10 secret exclusions + §11.4.28 owned-submodule paths; .codegraph/codegraph.db gitignored; §11.4.77 regen manifest + scripts/codegraph_reindex.sh. MCP wired for all 5 agents (Claude Code .mcp.json NEW; OpenCode + Kimi audited; Crush .crush.json NEW; Qwen Code .qwen/settings.json NEW). All configs reference bare codegraph on PATH. Comprehensive docs/codegraph/README.md. Tests 20/21/22, challenges CH-20/21/22, mutations M16/M17.
- **A19 verbatim covenant propagation.** §11.4 2026-04-28 user mandate now LITERALLY present in project CLAUDE.md + AGENTS.md + QWEN.md (was only in Constitution.md). Verify.sh Layer-1 gate added; test 19 PASS=7/0/0; CH-19; M15 (mutates a temp copy — real CLAUDE.md is never touched).
- **A20 AUDIT-1 fix.** M4/M5 meta-test mutations were silently SKIPping on Darwin for the wrong reason (BSD sed -i quirk). Real root cause: the mutations target Linux-only cgroup/systemd-run wrapper code unreachable on Darwin. Fix: explicit uname -s topology guard per §11.4.3 + portable inplace_sed for Linux runs.
- **A21 AUDIT-2 fix.** tmx kill -t NAME (documented friendly verb) was ambiguous because tmux only knows kill-pane/kill-server/kill-session/kill-window. Wrapper now translates the bare kill verb to kill-session. Test 23 PASS=5/0/0; CH-23; M19.
- **Docs reorganised** under context-named subdirectories per the operator's invocation of the constitution rule: 7 docs moved (GUIDE/SCROLLING/CODEGRAPH/CONTAINERIZATION_PLAN/NATIVE_DUAL_OS_PLAN/PER_SESSION_ISOLATION_PLAN/PLAN_v1.0.4). Every reference in governance + scripts updated atomically.
- **§11.4.65 universal-Markdown export.** New scripts/export_docs.sh (pandoc HTML + weasyprint PDF, idempotent, per-file timeout 60s) refreshed siblings for every consumer Markdown.
- **§11.4.71 pre-push.** Parent + constitution/ + Containers/ all at upstream tip; no divergent commits.
- **Honest gaps logged (§11.4.6) — out of scope this cycle:**
 - Upstream constitution/QWEN.md covenant insert → separate PR to HelixDevelopment/HelixConstitution (operator forbids modifying constitution from inside this project).
 - Containers/QWEN.md create + populate → separate PR to vasic-digital/Containers per §11.4.28.
 - Shell parser for CodeGraph (currently only the C file indexed, 6 nodes) → upstream contribution to add tree-sitter shell.
 - Agent-driven unforgeable-challenge end-to-end test → classified AUTONOMOUS_DEIGNED per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only` PASS=18/0/SKIP=5, meta-test 26/0/SKIP=6 (all SKIPs §11.4.3 topology-correct), e2e 9/0/0.

§3.9 — Scrolling fix + HelixConstitution inheritance (2026-05-21)

Status: COMPLETE (2026-05-21).

Triggering event: operator research note — tmux scrolling (up/down of terminal output, especially inside the Claude Code TUI) did not work; requirement to scroll vertically from ANY computer or mobile phone (Termux/Android). Plus the mandate to incorporate the HelixConstitution submodule as the root of the Constitution / CLAUDE.md / AGENTS.md, inherited further.

- **A16 scrolling fix** (Fixed.md A16): `scripts/tmux.conf.template` — history-limit 2000→50000, mode-keys vi, WheelUp/WheelDown overridden to always drive copy-mode scrollbar (works on desktop mouse + trackpad + Termux touch-scroll), allow-passthrough + extended-keys + terminal-features extkeys for the Claude Code TUI, OS-adaptive @clip clipboard. NEW test 17 (operator-path, PASS=13/0/0), `verify.sh` Layer-1 static gate, TMUX-CH-17, M12 + M13 paired mutations.
- **A17 HelixConstitution** (Fixed.md A17): added the `constitution/` submodule (pinned 7f738df). Full refactor of `Constitution.md` to the extends-template form (Project Articles §101–§109); CLAUDE.md / AGENTS.md inheritance pointers; new QWEN.md. NEW test 18 (PASS=10/0/0), TMUX-CH-18, M14 + CM-CONSTITUTION-INHERITANCE paired mutations. The constitution mutation runs on a TEMP COPY — the real, decoupled `constitution/` submodule is never modified (operator directive).
- **Containers submodule:** adopted remote 4ca5491, which already carries HelixConstitution recursive inheritance (`find_constitution.sh`, QWEN.md, all four governance docs). Parent gitlink bumped b077f2c → 4ca5491.
- **Meta-test portability:** M1/M2/M3/M6 converted from GNU sed -i to a portable `inplace_sed` helper — they now run on macOS instead of silently skipping. Meta-test: 18 caught / 0 escaped / 6 skipped (was 10/0/10).
- Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --rebuild` GREEN, suite PASS=13/0/SKIP=5, meta-test 18/0/6, e2e 9/0/0.

§4 — Recent commits

- (this commit, 2026-05-21) — **A16 scrolling fix + A17 HelixConstitution inheritance (v1.0.3 / versionCode 4)**. `tmux.conf.template`: history-limit 50000, vi copy-mode, WheelUp/Down → copy-mode scrollbar override, Claude Code passthrough, OS-adaptive clipboard. HelixConstitution submodule added at `constitution/`; `Constitution.md` refactored to the extends-template form (§101–§109); CLAUDE/AGENTS inheritance pointers + new QWEN.md. Tests 17 + 18, challenges CH-17/18, mutations M12–M14 + CM-CONSTITUTION-INHERITANCE; meta-test M1/M2/M3/M6 made portable. Containers gitlink → 4ca5491. Gate: PASS=13/0/5, meta 18/0/6, e2e 9/0/0. See Fixed.md A16 + A17, §3.9.
- (this commit, 2026-05-16) — **A15 cosmetic .exe-strip cycle (v1.0.2 / versionCode 3)**. Bottom-left status bar showed `claude.exe` because Claude Code v2.x ships its macOS native binary literally as `.../@anthropic-ai/claude-code/bin/claude.exe` (verified Mach-O 64-bit ARM64). Patched `scripts/tmux.conf.template` to set `automatic-rename-format "{s/\\.exe$/:pane_current_command}"` — literal-dot-anchored, escape-verified-empirically. Added test 16 (operator-path per §11.4.7, includes regression guard binary `t16_bashexe` to catch the unescaped-dot bug class), challenge TMUX-CH-16, M11 paired mutation. Live operator-path validation

recorded: `pane_current_command='claude.exe' → #W='claude'`. Full gate: PASS=11 FAIL=0 SKIP=5 (same SKIP profile as v1.0.0). See Fixed.md A15.

- (commit before that, 2026-05-13) — Audit cycle: tmux submodule reverted from 3f651d9f to cc117b5 (3.6a tag) after silent pin-drift caught; README.md test-count + PASS/SKIP staleness fixed (8→14 tests, PASS=6→10, SKIP=2→4); CLAUDE.md + AGENTS.md gained project summary, cross-links, fresh-conversation workflow, NN_*.sh glob fix, named four layers, "Files to never edit directly"; Issues.md restored A/B/C/D/E section headers; CONTINUATION.md §3.8 + timestamp refresh per §12.10. See §3.8.
- f4132aa — Auto-commit: opaque submodule bumps (Containers e377dea→af51968 legitimate upstream governance; tmux cc117b5→3f651d9f reverted in this commit). §12.10 / §11.4.6 violation documented in §3.8.
- (previous commit) — §11.4.4 layer-4 paired-mutation harness (META-MUT-001): meta_test_false_positive_proof.sh with 5 mutations (10/0/0 all caught); CLAUDE.md/ AGENTS.md PENDING→LANDED; Issues.md now empty of open items. Also: fixed 5 broken script references (REPO_ROOT wrong depth, filenames); implemented systemd-run --user --scope wrapping in tmx.template; test 09 topology dispatch fixed (functional probe, not mount grep); CLAUDE.md compacted to match AGENTS.md; binary built and verified GREEN (PASS=10 FAIL=0 SKIP=4: OOM helper + 3 destructive).
- 39284ab — Full coverage: CH-09 added (was missing); tests 12/13/14 (T5/T7/T8 destructive) with TMX_TEST_DESTRUCTIVE=1 gate; topology probe in tmx.template; challenge entries CH-12/13/14; B1/C1/C2/C3/D1 migrated Issues→Fixed.
- 8b37046 — Hostname-derived status-bar colour: DJB2→27-colour palette algorithm + wrapper integration + tests 10/11 + challenges + docs.
- 68a65b0 — Anti-bluff enforcement: TEST-AUDIT-001 complete (9 tests §11.4.2-audited), challenges paths fixed, AGENTS.md compact rewrite, covenant propagation verified.
- b92bf7f (2026-05-08) — §1 anti-bluff covenant verbatim user-mandate quote added; test 09 crash-isolation-scope (14/0/0) landed.
- 08d4ba5 (2026-05-07) — Initial vasic-digital/tmux: migrated from ATMOSphere project + per-session containerization plan + 8-test verification gate.

§8 — Resumption recipe

1. `cd ~/Projects/tmux`
2. Read this document
3. Read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md
4. Find the topmost IN PROGRESS / OPEN item, resume
5. Update this document in the SAME commit as the work itself (§12.10 invariant)